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Unit 5 : Session 9 : SLO 2 | Instruction Encoding and Addressing Modes

I. Multiple Choice

1. B) Represent instructions in binary.
2. B) Fixed-length encoding RISC's uniform length simplifies fetch/decode and pipelining.
3. C) How operands are accessed.

II. Fill in the Blanks

1. machine code encoding translates assembly into binary machine code.
2. fixed RISC uses fixed instruction length.
3. is accessed the addressing mode specifies how an operand is accessed.

III. True or False

1. TRUE decode complexity from encoding directly affects pipeline efficiency.
2. FALSE variable-length instructions complicate and slow decoding.
3. TRUE.

IV. Match the Following

- A) Opcode 2) Operation to perform.
B) Operand Specifier 3) Register/memory location info.
C) Addressing Mode 1) How the operand is accessed.
D) Instruction Length 4) Bits used per instruction.

V. Numerical Answers

1. 128 $2^7 = 128$ possible opcodes with a 7-bit opcode field.
2. 26 bits $32 - 6 = 26$ bits remain for operands/mode fields.
3. 32 $2^5 = 32$ registers are addressable with 5-bit specifiers.